

AI Music Generation and the Creative Process

Project Overview:

The goal of this project was to explore the intersection of artificial intelligence (AI) and music production by leveraging a variety of AI-driven tools to compose, arrange, and produce an original track. The project involved collaboration with six different AI technologies, each serving a specific purpose in the process of generating, refining, and finalizing a song. Below is a detailed breakdown of the process, highlighting the functions and capabilities of the AI tools utilized.

Process Overview:

Idea Generation with Udio: To begin the project, I used *Udio*, an AI-driven music composition tool, to generate initial song ideas. Udio allows users to input descriptive keywords and generate musical compositions based on those prompts. For this project, I provided keywords such as “Math rock,” “arpeggiated guitars,” “coastal,” and “love.” These keywords were chosen to evoke a certain atmosphere and musical texture, blending complex rhythms and instrumental colors with emotional undertones. The AI system produced a track entitled *Coastal Reflections*, which resonated with me both musically and thematically. This track formed the basis of my project, and I have linked it below for reference.

1. Udio - Cosmic Reflections.mp3

Stem Separation with Fadr: After selecting the generated track, I used an AI-powered stem separation tool from *Fadr* to isolate the individual elements (or stems) of the composition. Fadr’s technology utilizes deep learning algorithms to dissect a full mix into its constituent parts, such as vocals, drums, bass, and instrumental sections. This process not only allowed me to manipulate the isolated stems but also enabled me to determine the key and BPM (beats per minute) of the track—critical information for further arrangement and production.



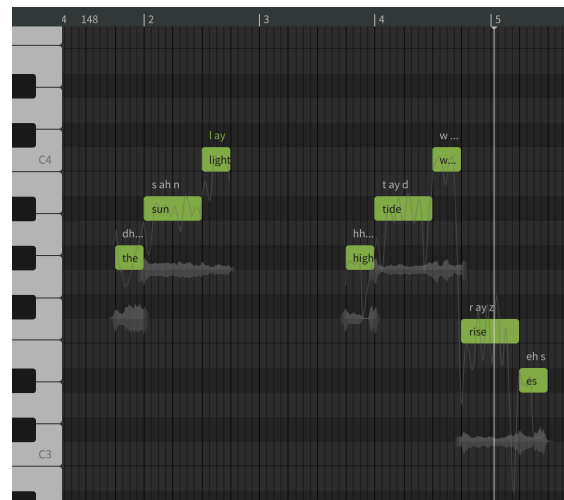
Arrangement in Ableton Live: With the separated stems in hand, I imported them into *Ableton Live*, a digital audio workstation (DAW) commonly used for music production. Using Ableton's intuitive interface, I created the structure for a two-to-three-minute song, arranging the stems to form an engaging and cohesive composition. I added my own bass and atmospheric pad sounds to the track, enriching the arrangement with additional layers and textures. At this stage, I focused on building the foundational elements of the song, such as its chord progression, rhythm, and overall sonic direction.

2. Cosmic Flip Structure.mp3

Lyric and Melody Generation with Suno: After finalizing the instrumental structure, I decided to integrate vocals into the track. I used *Suno*, an AI tool designed for generating lyrics and melodies based on uploaded audio. By uploading the instrumental track to Suno's platform, I was able to generate a full set of lyrics and a melody that complemented the existing musical arrangement. The AI-generated melody for the chorus was particularly fitting and inspired me to incorporate it into the final version of the track. However, I felt that the verse melody lacked the personal touch I desired, so I decided to create my own melody for the verses, maintaining a balance between human creativity and AI assistance.

3. Suno - Unseen Threads.mp3

Vocal Synthesis with Synthesizer V: With the melody and lyrics established, I turned to *Synthesizer V*, an advanced AI vocal synthesis tool that enables the creation of realistic vocal performances. Using Synthesizer V's MIDI input capabilities, I composed the vocal lines by inputting the melody as MIDI notes and typing in the lyrics. Synthesizer V offers a high degree of control over vocal phrasing, pitch, and expression, allowing me to craft a vocal performance that aligned with my vision for the track. Additionally, I used Synthesizer V to generate harmonies and vocal doubles, enriching the overall vocal arrangement and providing depth to the performance.



4. Synth V - Cosmic Flip.mp3

Vocal Transformation with Audimee: To further refine the vocal sound, I utilized *Audimee*, an AI-based vocal transformer that specializes in altering the tonality and character of synthetic voices. Using Audimee, I adjusted the vocal performance to achieve a softer, more intimate tone, which better suited the emotional atmosphere of the track. This transformation was crucial in ensuring the vocals blended seamlessly with the instrumental elements, creating a cohesive and emotive sound.

Jordan

Jordan's voice model is characterized by its deep, resonant tones, emotive delivery, and distinctive timbre. It embodies a deeply soulful quality, blending elements of R&B, soul, and alternative influences. The voice offers a rich and atmospheric sound, often conveying a melancholic undertone.



RnB Soul Alternative Unprocessed Emotive Melancholic

Jordan



5. AudiMee - Morning Tide.mp3

Final Mixing and Mastering with BandLab: After completing the arrangement, vocals, and additional production elements, I focused on fine-tuning the track through mixing. I made subtle adjustments to levels, panning, and effects to ensure a balanced and polished sound. For the final step, I used *BandLab*, an online music production platform that offers AI-powered mastering tools. BandLab's mastering service analyzed the track and applied automated adjustments to optimize the overall loudness, EQ, and dynamic range, ensuring the song met professional standards for release.

Morning Tide Mix1.wav Loop Master New Track

1:18 2:49

Original **Mastered** Shortcuts Reset All Intensity

- Universal**
Natural dynamic and tonal balancing.
- Fire**
Punchy lows and midrange clarity.
- Clarity**
Pristine highs with light dynamic expansion.
- Tape**
Warm saturation with analog dynamics.
- Natural**
Balanced dynamics and gentle compression.
- Spatial**
Atmospheric reverb and enhanced stereo width.
- Cinematic**
Intense saturation and harmonic distortion.
- Punch**
Energetic bass combined with boosted highs.

Download

Conclusion:

This project demonstrates the potential of AI tools in the music production process, showcasing how artificial intelligence can aid in various stages of music creation, from idea generation to final mastering. Each AI technology employed in this project brought unique capabilities to the table, allowing me to enhance my creative process while experimenting with new forms of collaboration between human and machine. By combining the strengths of AI with traditional music production techniques, I was able to create a track that reflects both innovation and artistic expression.

FINAL SUBMISSION:

 Final Submission - Morning Tide.wav

Lyrics:

The sunlight. High tide when it rises.

The night sky. Lights up as we dive in.

And I don't wanna waste a second of this time.

And for the first time I've been living right in line.

Love like fire can't be tamed

Burns so bright but it can fade

The night falls. Moonlight paints us silver.

The sea calls. Every wave pulls us nearer.

And I don't wanna trade this rhythm for the past

With every breath that i am given home at last

Love like fire can't be tamed

Burns so bright but it can fade

We are the stars who tried to collide

Now I'm washing it off in the morning tide

Love like fire can't be tamed (in the morning tide)

Burns so bright but it can fade (in the morning tide)

Love like fire can't be tamed (in the morning tide)

Burns so bright but it can fade (in the morning tide)

Love like fire can't be tamed

Burns so bright but it can fade

We are the stars who tried to collide

Now I'm washing it off in the morning tide